



Final-Synopsis

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Exploring Data Analytic Models for Household Food Insecurity Prediction

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Contents

1	Introduction	2
1.1	Status of Household Food Insecurity in Ethiopia	3
1.2	Household Food Insecurity status Data	4
1.3	Predicting Household Food Insecurity	5
1.4	Literature Review	6
1.5	Motivation	8
1.6	Research Objectives	9
2	Dataset Collection and Preprocessing	9
2.1	Data Preprocessing	9
2.1.1	Data Integration	9
2.1.2	Data Transformation	10
3	Feature Selection Methods for Predicting Household Food Insecurity Status	10
3.1	Filter Feature Selection Approach	11
3.1.1	Experiment and Results	14
3.2	Wrapper Feature Selection Approach	16
3.2.1	Experimental Results	17
4	Ensemble Learning Models for the Prediction of Household Food Insecurity Status	19
4.1	Experiment and Results	20
5	A Comparison of Machine Learning and Deep Learning Models for Predicting Household Food Insecurity Status	22
5.1	Experiment and Results	22
6	Conclusion and Future works	23
7	Publications	24

Exploring Data Analytic Models for Household Food Insecurity Prediction

Abstract

The need to conduct research in the field of household food security helps to understand the current status of household food insecurity, design and implement intervention programs. It helps to answer several important and critical questions by the extraction of useful knowledge that enables support for cost savings and decision making. Moreover, it helps domain experts to understand who is food insecure and secure and hence, makes decisions regarding distribution of food supplies. Predicting the status of food security is critical in presenting the household's situation to the appropriate agencies, which are in charge of taking preventative and intervention measures. The Ethiopian Household Income, Consumption, and Expenditure (HICE) survey data have created a large amount of data that can be used for study in various areas. The HICE dataset has a large amount of data that has not been analyzed. The main objective of this work is to develop predictive models for household food insecurity prediction on HICE dataset using ML and DL models. HICE contains 58,064 instances, of those, 32,378 instances belong to 'secure' class, and the remaining 25,686 to 'insecure' class. The dataset has 21 variables including the target class. Some of the variables are numeric, while others are nominal. The study employed, filter and wrapper-based features selection approaches to improve the classification accuracy.

To evaluate the models' performance on the HICE dataset, several ML Algorithms, namely: k-Nearest Neighbor (kNN), Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), and Navie Bayes (NB) are used. In addition, voting ensemble learning methods (hard and soft voting) are also employed to predict the household food insecurity status. Finally, DL models such as Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) are also applied to predict household food insecurity status. The standard evaluation measures such

as accuracy, precision, recall, F1-score, and ROC are used to evaluate the models' predictive performance. The experimental result, shows that the DL models outperformed compared to ML classifiers.

1 Introduction

The world in 21st century is witnessing considerable progress in industry, economics, and finance. Significant progress is also made in boosting health and lengthening the lifespans. According to Food and Agriculture Organization (FAO), despite the improvement in boosting health, more than a sixth of the world's population suffers from chronic hunger and malnutrition due to food insecurity. Food insecurity is defined as a lack of access by all people to enough food for active and healthy life (World Bank, 1986) and food security is described as the availability of sufficient global food supply of vital commodities to ensure a steady rise in food consumption while balancing variations in production and cost [17]. Food security is also defined in terms of the minimum cost of food that would meet the average nutritional needs of families of different sizes and compositions. Food insecurity is also defined as the lack of secure access to people for sufficient amounts of safe and nutritious food for average growth, development, and healthy life. The unavailability of food, insufficient purchasing power, and inappropriate food distribution at the household level may affect the people in various ways [12].

Food insecurity is a concern in any society that is growing and becoming a significant problem due to manufactured wars, climate change, and economic downturns. A household is likely to suffer from food insecurity if its total earnings is less than the food poverty line. According to the United States Department of Agriculture (USDA), accurate monitoring of food insecurity is essential for successfully operating federal government programs to assist food insecure households, food assistance programs, and other government initiatives to reduce food insecurity [9]. In the year 2020, around 2.37 billion people in the globe did not have access to sufficient food, an increase of over 320 million individuals in just one year [21]. Many individuals have been malnourished for at least the last 15 years

[3]. Despite significant reduction in the proportion of undernourished persons, from 69% in 1994/95 to 35% in 2013/14 (FAO-FSI, 2014), the number still remains high. In 2018, it was estimated that more than 821 million individuals throughout the globe were food insecure [20]. Food insecurity problems are becoming more frequent and severe in many places around the globe. Hence, predicting the state of food insecurity is essential to take early intervention action for the distribution of food by humanitarian organizations [24] in case of emergencies. Faster action in times of food crisis may save lives and the resources as well.

1.1 Status of Household Food Insecurity in Ethiopia

No countries in the world have achieved total food security. However, the problem is worse in sub-Saharan Africa. In Africa, around 204 million individuals were identified as malnourished between 2000 and 2002, with 86 million from the east, including Somalia, Ethiopia, Sudan, Kenya, and Tanzania [14]. Food insecurity has been a significant source of worry in Ethiopia [3]. According to the FAO's 2010 State of Food Insecurity report, around one billion people will be food insecure by 2050, accounting for 98% of developing nations [1].

For many years, Ethiopia has been troubled by food insecurity. According to the Ministry of Finance and Economic Cooperation (MoFEC), the average yearly income in Ethiopia is about \$883, and this makes the country among the world's poorest [22], and it has a significant population believed to be living below the poverty line [10]. According to the Global Hunger Index (GHI), Ethiopia ranks 90 among 116 world's hungriest countries [16].

The government, the World Food Program (WFP), and other organisations are working hard to help end the hunger crisis in Ethiopia. Conflict, drought, floods, locust invasions, market disruptions, high food prices, and the COVID-19 pandemic have made about 13.6 million people unable to get enough food [16]. As a result, Ethiopia has suffered through extended times of food insecurity. More than half of the population is at the

risk of starving because rural areas have fewer resources to grow and purchase food (Sisay Adugna, 2001). As a result of both natural and human-caused factors, food insecurity in Ethiopia is on the rise, putting an estimated five to six million people at risk of chronic and temporary hunger each year, with a further ten million people having their resilience compromised as a result [5].

1.2 Household Food Insecurity status Data

The Household Income, Consumption, and Expenditure survey, also known as the Household Income and Expenditure Surveys (HIES), the Household Budget Survey (HBS), or the Living Standards Measurement Survey (LSMS), is a complex survey done on a nationally representative sample to describe important aspects of household socioeconomic conditions, such as how they get and/or use food. HICE surveys are usually done once every three to five years and cover 7,000 to 20,000 households to get a statistically representative sample.[11]. Most HICE surveys are conducted by national statistical agencies, with the technical assistance from the World Bank’s Living Standard Measurement Study (LSMS) group. It primarily reflects the income aspect of the household and gives information on households’ income, consumption, and expenditures. The core objective of the HICE survey is to provide data that helps to understand the income aspects of population. The primary objectives are to:

- evaluate the level, extent, and distribution of income in poverty
- provide information on the levels, distribution, and household expenditure patterns to study how households’ living standards have changed over time in different socioeconomic groups and geographic areas.
- provide basic data to design, monitor, and evaluate the effects of socio-economic policies and programmes on household living standards.
- provide a set of data to evaluate poverty levels in general and food security in particular.

- provide information for the system of national accounts compilation of household accounts, especially for private consumption expenditure evaluation, and
- collect data to develop consumer price indices at different levels and in different regions, such as weights.

The data in the HICE survey is categorized into 'food secure' and 'food insecure' class based on income of the households. Hence, this survey data can be used, to conduct implement to predict the insecurity status of households. The Ethiopian HICE survey is conducted by the Ministry of Finance and the Central Statistical Agency (CSA), once in every five years. The data collected from the people starting in 9 regional states and two city administrations of Ethiopia in 2011, and 2016 is used to construct the HICE dataset to predict the household food insecurity status. Of the 58,064 records, 27,834 instances are from the 2011 survey data, whereas the remaining 30,230 are from the 2016 survey data.

1.3 Predicting Household Food Insecurity

Food aid and humanitarian assistance cannot be provided unless the extent and breadth of food insecurity emergencies are accurately and appropriately identified. On the other hand, policymakers often lack the knowledge of people for whom the food aid has to be distributed [26]. If food insecurity disaster is forecasted early, stakeholders can take necessary steps to prevent or control its impact. This strategy has been successful in some areas of the world [9]. Various studies have highlighted the importance of predicting or monitoring food insecurity status based on household survey data, using a wide range of Machine Learning (ML) and Deep Learning (DL) technologies [7] [19]. Predicting household food insecurity status can be modeled as a typical binary classification problem with the class labels "food secure" and "food insecure." ML algorithms, namely: Support Vector Machine (SVM), Random Forest (RF), Gradient Boosting (GB), and Decision Tree (DT), are popularly used to predict the status of household food insecurity. Off-late DL algorithms are also becoming popular for predicting food insecurity status.

1.4 Literature Review

Several researchers have explored the potential issues in predicting the household food insecurity status. The following is a summary of notable research discoveries made in the area of predicting household food insecurity status is as follows:

FH Bitew et.al. [6] used ML algorithms, including eXtreme GB, k-Nearest Neighbor (kNN), RF, Artificial Neural Network (ANN), and generalized linear models on the Ethiopian Demographic and Health Survey (EDHS) data of 2016 consisting of 9,471 children information who are below five years. The eXtreme GB algorithm outperformed other ML algorithms for predicting childhood food insecurity with an accuracy of 86.8%, and the findings obtained support the improved access to water supply, food security, and fertility regulation in Ethiopia's quest to improve food insecurity. Barbosa, R. M., Nelson, D. R. [4] used a two-stage stratified random sampling to collect household survey data from 80 households belonging to six distinct municipalities amounting to 480 households. Out of 75 features, they used 14 features to train an SVM classifier to classify household data into food secure and insecure categories and obtained 77% accuracy. Gao, et al. [13] identified household indicators that separate food-insecure families from food-secure ones, allowing for more accurate targeted aid to the latter, using the 2008 Afghanistan National Risk and Vulnerability Assessment (NRVA) survey data that included 20,511 randomly selected households nationwide. They trained DT and RF models to predict food secure and food-insecure families, and the RF model obtained an accuracy of 80%. The results demonstrate that the six variables: income and expenditure items, household size, farm-related measures, access to particular resources, short-term shocks, and dwelling wall composition are the important determinants of food security level. Research conducted by Okori et al. [19] for the prediction of food security used Uganda Bureau of Statistics secondary data for two agricultural seasons on agricultural households in four regions of Uganda. The dataset comprise 3,030 instances with 24 features, though the authors used only 15 features in their experiments, as the remaining features were either transformed or derived from other features. ML algorithms namely: SVM, kNN, Navie Bayes (NB),

and DT were applied to predict food insecurity status, and SVM and kNN outperformed the other classifiers.

Van der Heijden, Wesley, et al. [23] proposed a method for predicting food insecurity in Ethiopia at the sub-national level using open data linked to food insecurity drivers, and the RF model obtained an accuracy of 90%. Nigus, M. [18] used GB, RF, Extra tree (ET), Bagging, kNN, DT, SVM, LR, Ada-Boost (AB), and NB classification models to predict household food security status. The authors conducted the experiment on Ethiopian HICE survey data and obtained 99.97% accuracies using RF and GB classifiers. M. Align et al. [2] employed RF, SVM and DT classifiers to identify risk indicators and predict food security status at the household level. On Dabat Health and Demographic Surveillance (DHPS) data, the experimental findings indicate that the DT algorithm achieved an accuracy of 91.58%.

Meerza and A. Ahamed [15] used Bangladesh’s Integrated Household Survey (BIHS) data, the most comprehensive nationally representative rural household data, to obtain household-level data on the dynamics of poverty, food security, and agricultural development in Bangladesh. This survey tracks household consumption of 321 food items during seven days. The authors utilised a food consumption table to convert food intake into calories and estimated calorie intake per family member each day by dividing household calorie intake by household size. Based on age and gender, they predicted 2,122 calories per household member each day. Food insecure households have a daily calorie intake of less than 2,122. According to BIHS data, while 4,461 households are food insecure, the remaining 1,955 are food secure. Experiments were conducted using SVM, RF, and LR classifiers to predict household food security status and RF, SVM, and LR classifiers obtained accuracies of 83.3%, 71.1%, and 69.6%, respectively. Due to the limited dataset size (6,461 households) the results obtained are not encouraging.

Yujun Zhou and Kathy Baylis [26] predicted food security status based on high-frequency data received remotely that consists of information related to precipitation, temperature, market prices, soil quality, and geographic characteristics. The authors used Ensemble of RF and GB models to improve the accuracy of predictions, and obtained about an r

squared of 0.6–0.7 at the cluster level and about 0.4 at the household level. Several food security indicators established by international relief organizations, such as the Household Dietary Diversity Score (HDDS), reduced Coping Strategies Index (rCSI), and Food Consumption Score (FCS), were used by the authors to predict household food insecurity status.

1.5 Motivation

Predicting food insecurity status is critical in assisting stakeholders in implementing early intervention relief efforts. Policymakers, scholars, government and non-government groups, and development workers are paying close attention to the issue of food insecurity status. Predicting food insecurity in emerging nations, especially in African countries such as Ethiopia, is difficult because the required data is not readily available in the usable format. For this reason, governments do not have access to appropriate and relevant information for making decisions on the measures required to stabilize their countries' food security situation and status. Ethiopia's food security remains a top priority for the country's administration. The Ethiopian HICE dataset provide massive amounts of information about the characteristics of a household survey. This survey data is used by many academics, policymakers, and humanitarian organizations to analyze household food security status. Even though little research has been done on household food security status prediction using household survey data, it is observed that the unreliability of datasets is a challenge for predicting household food insecurity status.

To overcome the noted gaps, the research work focuses on developing ML and DL models to predict household food security status. The outcome of this work is expected to help the policymakers, scholars, government and non-government groups, and humanitarian organizations to make vital decisions and intervention actions in distributing the food supplies.

1.6 Research Objectives

The following list of objectives are explored for the prediction of household food security status:

1. Feature selection techniques to select the significant features
2. Designing ML models for predicting household food insecurity status
3. Designing DL models for predicting household food insecurity status

2 Dataset Collection and Preprocessing

The Ethiopian HICE survey data corresponding to the years 2011 and 2016 is used for this work. The class distribution of instances reveals that 32,378 instances belong to the secure class, while the remaining 25,686 belong to the insecure class and there are 21 features, including the target class ("food secure" or "food insecure."). Some features are numerical, while others are nominal.

2.1 Data Preprocessing

Data preprocessing refers to cleaning raw data and to transform the data to a more manageable format so that it can be used by subsequent models for further processing. The preprocessing steps used in this work are data integration and data transformation.

2.1.1 Data Integration

Data integration combines data from several sources to create a more comprehensive picture. HICE survey datasets of 2011 and 2016 are integrated to form one dataset using the SPSS tool¹. Of the 58,064 records, in the integrated dataset, 27,834 instances are from the 2011 survey data, and the remaining 30,230 are from the 2016 survey data. The survey data of both the years contain 21 features.

¹<https://spss.en.softonic.com/download>

2.1.2 Data Transformation

Data transformation is the conversion of data from one format to another, as per the requirement. The data transformation methods include smoothing, integrating, generalizing, normalizing, discretion, and attribute generation [25]. The HICE dataset available in .SAV format is converted to .CSV format as is easy to use.

3 Feature Selection Methods for Predicting Household Food Insecurity Status

A large amount of data which is widely available is attracting the researchers to turn it into useful knowledge, but real-world data may be low quality and may contain unreliable and redundant data, which may affect the performance of data analysis. Hence, there is a need for relevant and unredundant data. Feature selection is the process of selecting the most important features and removing the less important ones. The fundamental goal of this work is to select features for the accurate prediction of household food security status using ML approaches. Filter and wrapper-based feature selection strategies are employed to select the relevant features from the given dataset. ML classifiers, namely: Random Forest (RF), k-Nearest Neighbor (kNN), Support Vector Machine (SVM), Logistic Regression (LR), and Naive Bayes (NB) are employed to evaluate the performance of the selected features. Figure 1 shows the ML framework for predicting the household food insecurity status.

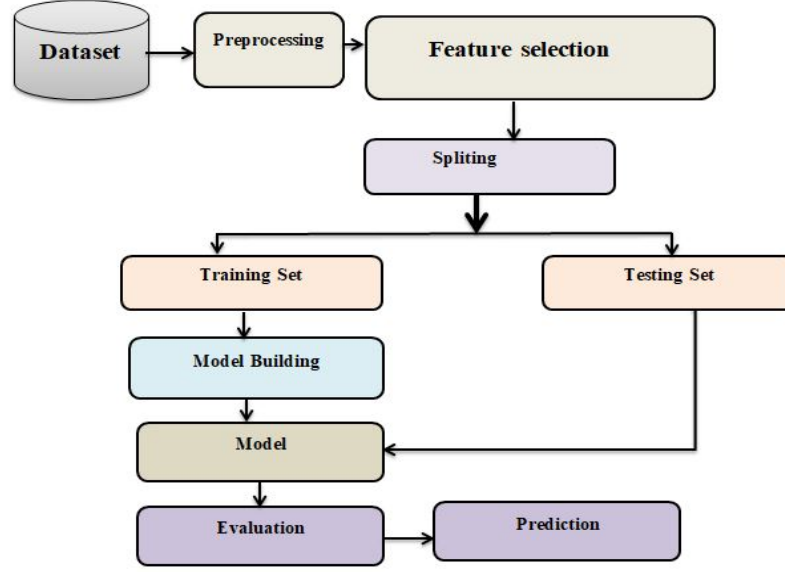


Figure 1: Machine Learning framework for predicting the household food security status

3.1 Filter Feature Selection Approach

Filter methods use statistical approaches to evaluate the relationship between each input variable and the target variable, and these scores are used as a foundation for selecting the features [8]. Filter methods are frequently used for feature selection because they are usually faster and independent of any ML algorithm. A feature significance score is calculated based on its relation with the target class, and the features are ranked based on the significance scores. Further, top 'k' of these features can be selected based on the value of 'k'. Feature selection strategies, namely: Chi-square, Feature Importance, and Correlation Coefficients are the focus of this work. ML classifiers, namely: RF, kNN, SVM, LR, and NB are employed to evaluate the performance of the selected features. Performance evaluation metrics, namely: accuracy, precision, recall, F1-measure, and ROC are used to evaluate the performances of ML classifiers on the HICE dataset.

- **Chi-square:** is a statistical test that measures a feature's independence from the class labels. It tests each attribute individually to determine the intensity of the relation between the feature and the target class. Because Chi-square metrics are model agnostics, they rely entirely on features and are computationally very fast. The main reason for selecting these Chi-square tests is that the HICE dataset consists

of discrete and categorical data that the Chi-square test supports. Table 3 shows the top 10 features selected using the Chi-square feature selection method.

Table 1: Top 10 features selected using Chi-square method

Sl.No.	Feature Name
1	Net calorie
2	Annual Expenditure
3	Weight
4	Household head age
5	Kebele
6	Household size
7	Report level
8	Kefeke ketema
9	ADEQUIV
10	Region

- **Feature Importance:** is a strategy that calculates a score for feature, and the relevance of a feature is measured in terms of the information gained compared to the target class. Higher the score, more significant the feature is. Extra Trees (ET) Classifier is used to select the most important features with respect to target class. Figure 2 shows the top 10 features selected using the Feature Importance selection technique.

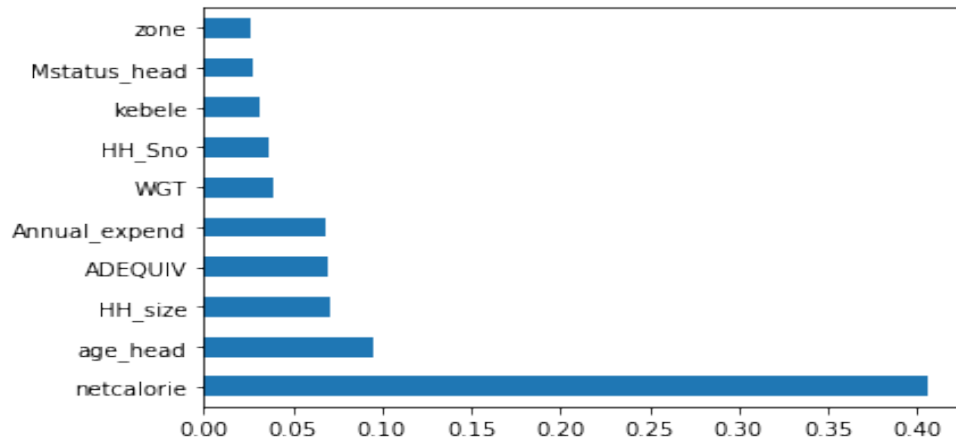


Figure 2: Top 10 features selected using feature importance method

- **Correlation Coefficient:** This approach tests whether the features are strongly correlated with the target class but do not correlate with each other. The correlation

coefficient assesses the features by considering each feature's potential prediction and the redundancy level. The correlation coefficient is represented with a heat map, which makes it easy to identify which features are most closely related to the target class. A heat map is a depiction of data that is two-dimensional and uses colours to represent the values of the data. Figure 3 shows the Correlation between features and target class.



Figure 3: Correlation between features and target class using Heatmap

Calculating the correlation coefficient based on the degree to which each feature is correlated with the target class and establishing a threshold value are the two steps in applying the correlation approach to choose the most valuable features. The features having a cor-

relation coefficient higher than the threshold are selected as relevant features.

3.1.1 Experiment and Results

The features selected using filter-based method, namely: Chi-square, Feature Importance, and Correlation Coefficient are used to train ML algorithms, namely: kNN, RF, SVM, LR, and NB with their default parameters and values to evaluate the performances of the features selected. The classifiers are evaluated taking top 10, 7 and 5 features and the performances of the classifiers, measured in terms of accuracy, precision, recall, and F1-measure are shown in the tables 2, 3, 4, and 5, respectively. The value of 'k' is set to five in kNN classifier.

Table 2: Accuracy of ML classifier with all features

Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-measure (%)
RF	99.96	99.96	99.97	99.96
kNN	94.28	94.7	93.82	94.35
LR	88.9	88.75	88.78	88.9
SVM	94.3	94.33	93.64	94.01
NB	76.04	76.40	76.68	76.10

Table 3: Accuracy of ML Classifiers using Chi-square feature selection

Classifier	Features selected	Accuracy (%)	Precision (%)	Recall (%)	F1-measure (%)
kNN	10	93	92	93	92
	7	95	95	94	94
	5	93	95	88	91
LR	10	89	87	88	88
	7	82	82	77	79
	5	90	91	87	89
RF	10	99.97	99.96	99.98	99.97
	7	99.93	99.89	99.96	99.92
	5	99.65	99.87	91	94.8
SVM	10	89	89	85	87
	7	88	88	86	87
	5	91	92	89	90
NB	10	77.7	71	83	76.7
	7	80	72	89	79.7
	5	77.7	68	88.6	76.9

Table 4: Accuracy of ML Classifiers using Feature Importance feature selection

Classifier	Features selected	Accuracy (%)	Precision (%)	Recall (%)	F1-measure (%)
kNN	10	91	87	92	93
	7	93	93	92	93
	5	96	97	95	96
LR	10	87	86	84	85
	7	89	87	87	87
	5	89	87	88	87
RF	10	99.98	99.96	100	99.98
	7	99.96	99.94	99.98	99.96
	5	99.96	99.94	99.98	99.94
SVM	10	96	96	95	95
	7	89	90	84	87
	5	89	90	84	87
NB	10	77	70	84	76.7
	7	78	71	82	76.9
	5	78.7	72	84	77.8

Table 5: Accuracy of ML Classifiers using Correlation Coefficient feature selection

Classifier	Selected Features	Accuracy (%)	Precision (%)	Recall (%)	F1-measure (%)
kNN	10	92	88	93	91
	7	95	95	94	94
	5	96	96	95	96
LR	10	89	87	88	87
	7	89	87	88	87
	5	89	87	88	87
RF	10	99.96	99.94	99.98	99.96
	7	99.97	99.96	99.98	99.97
	5	99.95	99.96	99.96	99.96
SVM	10	89	91	85	88
	7	90	88	88	88
	5	88	89	84	87
NB	10	77	70	81	75.9
	7	77	70	81	75.8
	5	78	72	84	77.8

As observed from the Tables 2, 3, 4 and 5, Feature Importance method is best for selecting features in HICE dataset compared to the other two methods.

3.2 Wrapper Feature Selection Approach

In wrapper methods, the selection of features is based on a specific ML algorithm that is fit on a given dataset. It uses a greedy search method that compares all possible combinations of features against the evaluation criteria. Lastly, it chooses the combination of features that gives the best results for the given ML algorithm. In this work, the most commonly used wrapper methods: Forward Feature Selection (FFS), Backward Feature Elimination (BFE), and Recursive Feature Elimination (RFE), are used to select the features. ML algorithms, namely: kNN, RF, SVM, LR, and NB, are used to evaluate the performance of selected features.

- **Forward Feature Selection:** method starts with zero feature and adds an important/relevant feature one at a time to the subset of already existing important features in each iteration. After each relevant feature is added to the subset of features, a ML algorithm is used to evaluate the features. The loop terminates when the subset of features does not improve the performance of the model further or all the features are exhausted. Random forest classifier is used to select the best features and relevant features are selected using the following process in the FFS method:

1. Choose a significance level.
2. Fit each model by examining one feature at a time, and select the feature with the smallest p-value.
3. Fit all of the models with an extra feature that wasn't chosen before.
4. Again, choose the feature whose p-value is the lowest. If the p-value is lower than the significance level, move on to Step 3.

- **Backward Feature Elimination:** method starts with all the features and eliminate an irrelevant/redundant feature one at a time from the subset of features in each iteration. After each irrelevant/redundant feature is removed from the subset of features, a ML algorithm is used to evaluate the features. The loop terminates

when the subset of features does not improve the performance the of the model further or no features are left out. Random forest classifier is used to select the best features. The BFE approach uses the following procedures for selecting irrelevant or duplicate features.

1. Choose a significance level.
 2. Build a model using all of the features.
 3. Consider the feature whose p-value is the highest. If the p-value is greater than the significance level, move on to Step 4. If not, stop.
 4. Eliminate the feature that is currently considered.
 5. Fit a model with the remaining features. Repeat the whole process from step 3.
- **Recursive Feature Elimination:** Starts with all features in the training dataset and successfully removes features until the desired number remains. This is achieved by fitting the given ML algorithm used in the core of the model, ranking features by importance, discarding the least important features, and re-fitting the model. This process is repeated until a specified number of features remains. When the full model is created, a feature importance measure is computed that ranks the predictors from most important to least. At each stage of the search, the least important features are iteratively eliminated prior to rebuilding the model. Recursive Feature Elimination Cross-validation (RFECV) method is used to determines the least important feature in each iteration. The RFECV method is implemented with parameters, namely: estimator=rfc, step=1, and cv=10-fold cross-validation.

3.2.1 Experimental Results

The features selected using FFS, BFE and RFE methods are used to train the ML algorithms, namely: kNN, RF, SVM LR, and NB to evaluate the performance of the features selected. The performances of the ML classifiers measured in terms of accuracy, precision,

recall, and F1-score are shown in Tables 6, 7, and 8, respectively. The value of 'k' is set to five in kNN classifier. The result shows that RF classifier scored best compared to other algorithms with an accuracy of 99.98% by using the features selected by BFE.

Table 6: Accuracy of ML Classifiers using Forward Feature Selection method

Classifier	Features selected	Accuracy (%)	Precision(%)	Recall (%)	F1-measure (%)
RF	10	99.5	99.94	99.96	99.95
	7	99.97	99.96	99.98	99.97
	5	99.96	99.92	100	99.96
LR	10	87	86	84	85
	7	88	86	87	87
	5	88	86	87	87
SVM	10	88	90	82	85
	7	91	93	85	89
	5	88	90	82	86
KNN	10	90	87	92	89
	7	92	91	92	91
	5	95	95	95	95
NB	10	80.11	87.25	75.34	80.86
	7	80.23	86.39	76.62	81.21
	5	81.21	87.95	76.84	80.02

Table 7: Accuracy of ML Classifiers using Backward Feature Elimination method

Classifier	Features selected	Accuracy (%)	Precision(%)	Recall (%)	F1-measure (%)
RF	10	99.98	99.94	100	99.97
	7	99.94	99.90	99.98	99.94
	5	99.96	99.92	100	99.96
LR	10	87	86	84	85
	7	88	86	88	87
	5	88	86	87	87
SVM	10	88	90	82	85
	7	91	93	85	89
	5	88	90	82	86
KNN	10	90	87	92	89
	7	92	91	92	91
	5	95	95	95	95
NB	10	80.19	86.50	76.39	81.13
	7	81.90	90.48	75.54	82.32
	5	81.50	88.91	76.36	82.16

Table 8: Accuracy of ML classifiers using Recursive Feature Elimination method

Classifier	Features selected	Accuracy (%)	Precision (%)	Recall (%)	F1-measure (%)
kNN	5	96	96	95	95
SVM		89	90	84	87
RF		99.94	99.96	99.92	99.94
LR		86	85	83	84
NB		81.13	89.68	74.77	81.55

4 Ensemble Learning Models for the Prediction of Household Food Insecurity Status

Ensemble methods are used to make learning models more accurate by combining more than one model instead of just using one. The base models are trained individually and the prediction on the test set are made based on the combined voting decision of the prediction of the base models. While the hard voting is based on simple majority voting, soft voting uses a weighted probability strategy. In this work, hard and soft voting classifiers are used to combine five base classifiers, namely: RF, kNN, SVM, Multi-Layer Perceptron (MLP), and Extra Tree (ET) for predicting the household food security status. Figure 4 depicts the proposed ensemble learning framework using hard and soft voting methods. Backward Feature Elimination method discussed in section 3 is used to select the features.

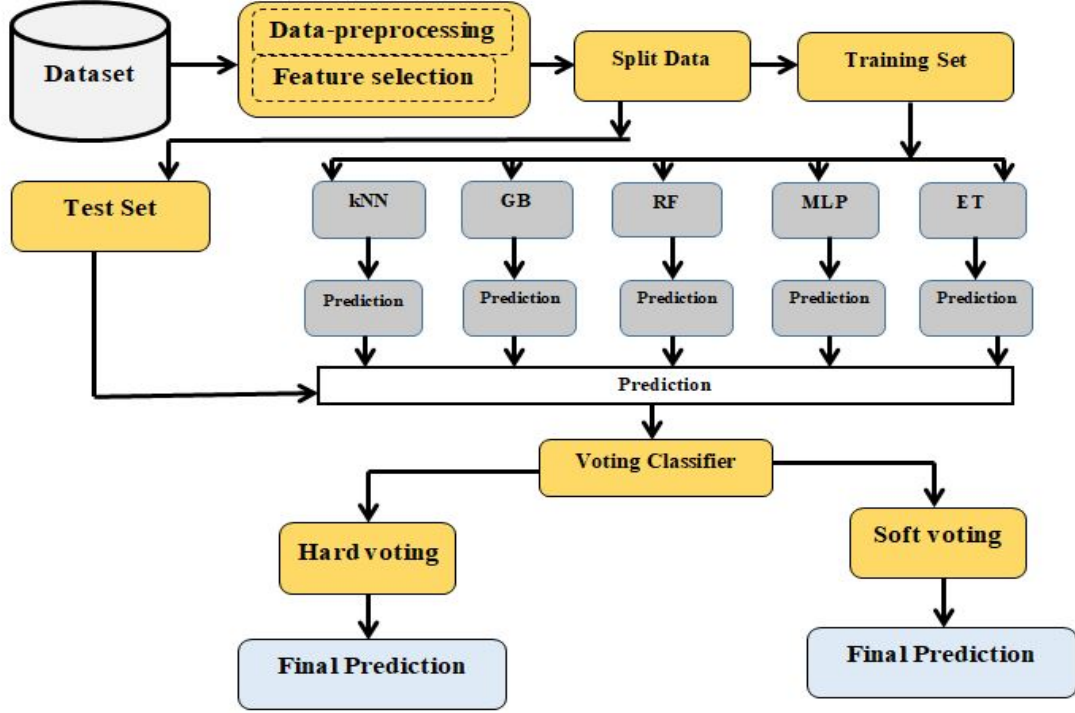


Figure 4: Proposed Ensemble Learning method

4.1 Experiment and Results

The performances of ensemble classifiers as well as the base classifiers before feature selection are shown in Table 9. The RF classifier surpasses the other base classifiers as well as ensemble classifiers with an accuracy of 99.98%.

Table 9: Accuracy of Ensemble Classifier before feature selection

Classifier	Performance Evaluation Metrics			
	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
GB	99.96	100	100	100
RF	99.98	100	100	100
MLP	95.50	96	96	96
kNN	94.43	95	94	94
ET	95.55	96	96	96
Hard Voting	99.51	100	100	100
Soft Voting	99.79	100	100	100

Table 10: Accuracy of Ensemble Classifier after feature selection

Classifier	Performance Evaluation Metrics			
	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
GB	99.96	100	100	100
RF	99.96	100	100	100
MLP	97.99	98	98	98
kNN	95.86	96	96	96
ET	97.82	98	98	98
Hard Voting	99.76	92	92	92
Soft Voting	99.77	97	97	97

The performances of base classifiers and ensemble classifiers after feature selection are shown in Table 10. The comparison of the performances of soft voting and hard voting classifier reveals that soft voting classifier perform better before feature selection and hard voting classifier perform better after feature selection with an accuracy of 99.79% and 99.76% respectively.

ROC curve measures a classifier's performance and shows true positives versus false positives. An ideal classifier has a ROC of 100% true positive and 0% false positives and is used to measure proper positive classifications by the rate of false positives. The comparison of the ROC curves of RF, kNN, SVM, MLP, and ET before and after feature selection using backward feature elimination method are shown in Figure 8.

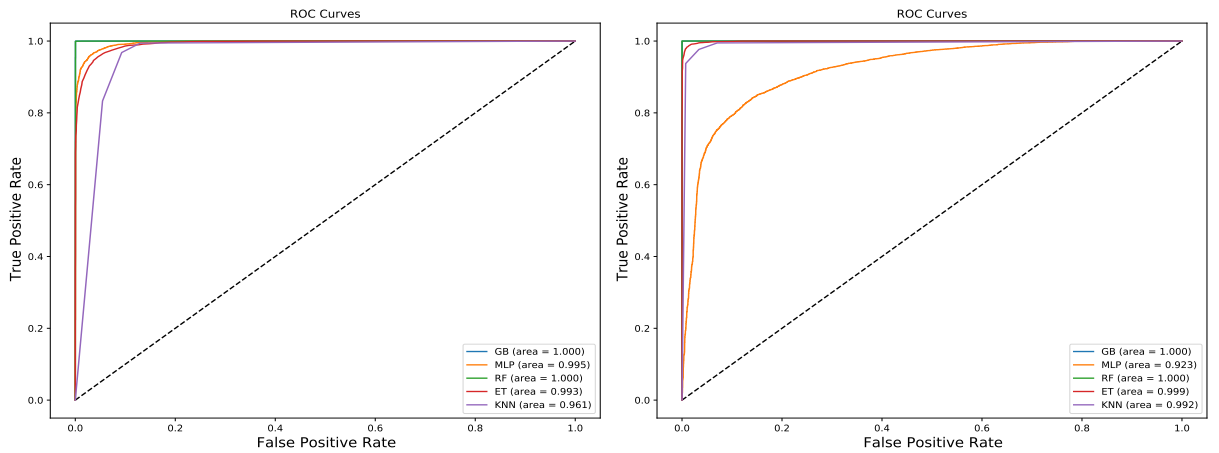


Figure 5: ROC results before feature selection (left), after feature selection (right)

5 A Comparison of Machine Learning and Deep Learning Models for Predicting Household Food Insecurity Status

ML and DL algorithms are becoming more popular to predict household food security status, which can be used by the governments and policymakers of the country to provide food supplies for the needy in case of emergency. ML models, namely: kNN, LR, SVM, NB, MLP and DL models, namely: Artificial Neural Network (ANN) and Convolutional Neural network (CNN) are applied to predict household food security status on Ethiopian HICE survey data. Figure 6 shows the DL framework to predict the household food security status. The standard evaluation measures, namely: accuracy, precision, recall, F1-score, and ROC are used to evaluate the models' predictive performance.

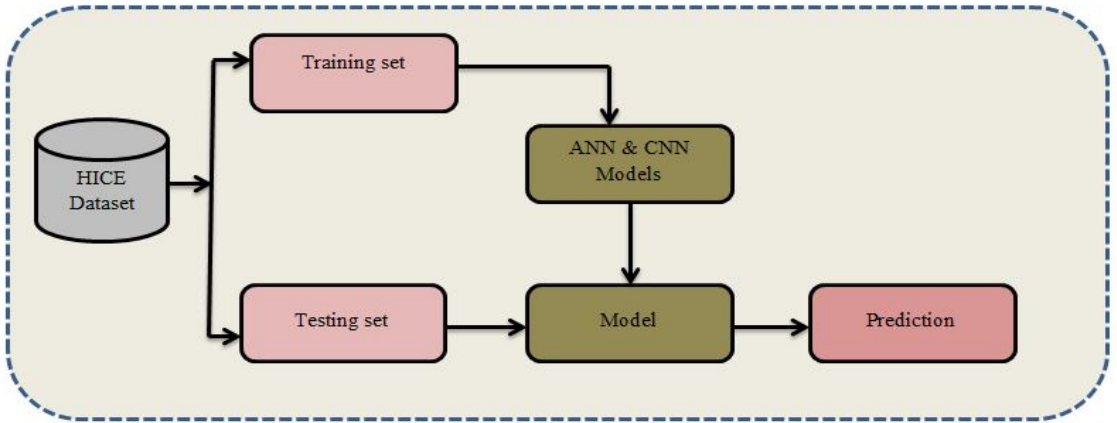


Figure 6: Deep Learning framework to predict household food security status

Backward feature elimination method is used to select the features for ML algorithm. However, all the features are used to train the DL models.

5.1 Experiment and Results

Figure 7 shows the comparison of the performances of ML and DL models. Among all ML and DL models, the DL model of ANN scores better than other classifiers with an accuracy of 99.51%. ROC curves of ML and DL models are shown in Figure 8.

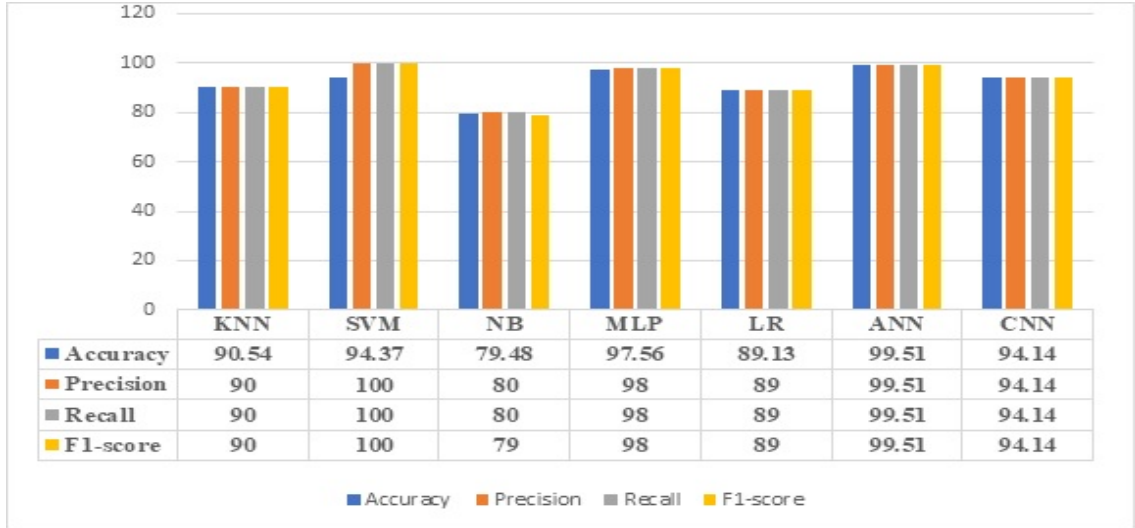


Figure 7: Comparison of the performance of ML and DL Models

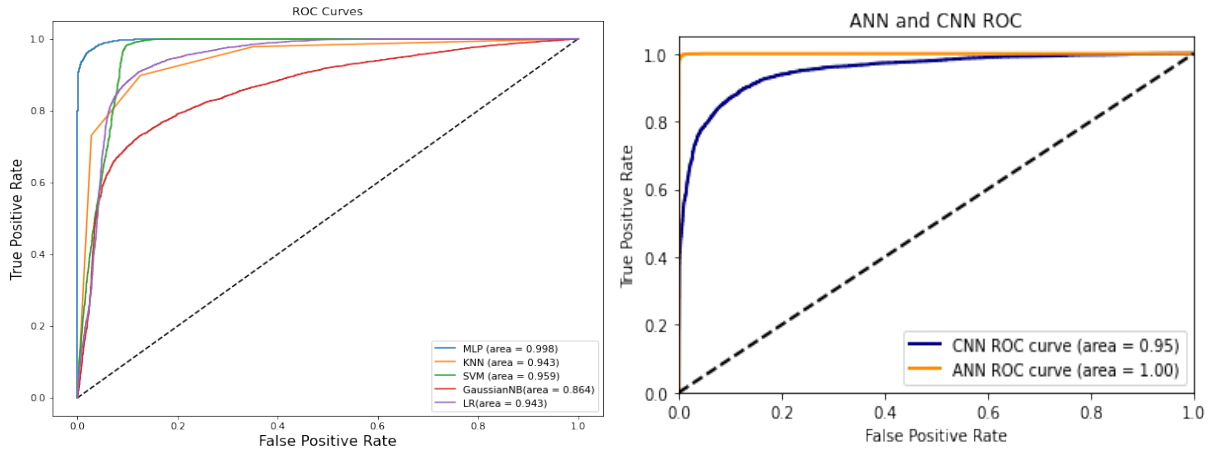


Figure 8: ROC curve of ML classifiers (left), ROC of ANN and CNN models (right)

6 Conclusion and Future works

Recently, a large amount of data widely available for ML has attracted researchers to turn such data into useful knowledge. Feature selection removes unnecessary and/or redundant features from the dataset to improve the performance of learning models. The filter and wrapper feature selection approaches are implemented in this work to improve the classification accuracy of ML models. ML algorithms, namely: RF, kNN, LR, SVM, and NB are applied to predict the household food security status. Further, Ensemble learning of RF, GB, MLP, kNN, and ET classifiers based on hard and soft voting are applied

to predict the household food insecurity status. The DL models, namely: ANN and CNN, are also investigated to predict the household food security status. All the learning models are evaluated on the Ethiopian HICE data set. Standard evaluation criteria such as accuracy, precision, recall, F1-score, and ROC are employed to evaluate the models' prediction ability. Experimental results show that ANN, a DL model, surpasses all the ML classifiers. The future directions include experimenting with different and large datasets as the dataset used in this work is small and belongs only to Ethiopia. Further, data such as land information, weather data, price data, conflict-related data, and so on will also be considered to predict household food security status.

7 Publications

1. Dorseywamy, Mersha Nigus, Feature Selection Methods for Predicting Household Food Insecurity *International Journal of Recent Technology and Engineering (IJRTE)*, 2020, pp.:1560-1568.
2. Mersha Nigus Alemayehu and Dorseywamy: Feature Selection for Household Insecurity Classification: Wrapper Approach *Asian Journal of Information Technology*, (ISSN:1993-5994), Volume-20 Issue-5, 2021 pp.146 - 151.
3. Mersha Nigus and Dorseywamy: Performance Evaluation of Classification Models for Household Income, Consumption and Expenditure Data Set *Asian Journal of Information Technology*, (ISSN:2320 – 8791), Volume-20 Issue-5, 2021 pp.134 - 140.
4. Mersha Nigus, Dorseywamy: A Study of Embedded Feature Selection Methods for Household Food Insecurity Classification on HICE Data *IJREAT International Journal of Research in Engineering Advanced Technology*, (ISSN:2320 – 8791)(Online), Volume 7, Issue 5, Oct - Nov, 2019 pp.6-11.
5. Dorseywamy, Mersha Nigus: Feature Selection Methods for Household Food Insecurity Classification 2020 International Conference on Computer Science, Engineering

and Applications (ICCSEA), 2020, pp. 1-7, IEEE,
DOI: 10.1109/ICC-SEA49143.2020.9132945.

6. Mersha Nigus and H.L Shashirekha: A Comparison of Machine Learning and Deep Learning Models for Predicting Household Food Security Status ”International Journal of Electrical and Electronics Research (IJEER”)
7. Mersha Nigus and H.L Shashirekha: A Comparative Study of Feature selection Methods on HICE Dataset, 2022 2nd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), 2022, pp. 2515-2521, IEEE, DOI: 10.1109/ICACITE53722.2022.9823585

Paper under Preparation

1. ”Ensemble Learning Models for the Prediction of Household Food Insecurity Status”

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